

Convex Optimization

HW #1

Exercise: 1

1) $S = \{x \in \mathbb{R}^n \mid d_i \leq x_i \leq \beta_i, i = 1, \dots, m\}$

$$S = \bigcap_{i=1}^m \left(\{x \in \mathbb{R}^n \mid -x_i \leq -d_i\} \cap \{x \in \mathbb{R}^n \mid x_i \leq \beta_i\} \right)$$

~~S~~ is a finite intersection of halfspaces. As a consequence, S is convex

2) $S = \{x \in \mathbb{R}^2 \mid x_1 x_2 \geq 1\}$

Let's $x = (x_1, x_2) \in S$ and $y = (y_1, y_2) \in S$ so $x_1 x_2 \geq 1$ and $y_1 y_2 \geq 1$

Let $\alpha \in [0, 1]$ and let's show that $\alpha x + (1-\alpha)y \in S$ which means

$$\underbrace{(\alpha x_1 + (1-\alpha)y_1)(\alpha x_2 + (1-\alpha)y_2)}_A \geq 1$$

$$A = \alpha^2 x_1 x_2 + \alpha(1-\alpha)y_2 x_1 + \alpha(1-\alpha)y_1 x_2 + (1-\alpha)^2 y_1 y_2$$

$$= \alpha^2 x_1 x_2 + (1-\alpha)^2 y_1 y_2 + \alpha(1-\alpha)(y_1 x_2 + y_2 x_1)$$

We have $y_1 y_2 \geq 1 \Rightarrow y_1 x_2 \geq \frac{x_2}{y_2}$ and $x_1 x_2 \geq 1 \Rightarrow x_1 y_2 \geq \frac{y_2}{x_2}$

$$\text{So } y_1 x_2 + x_1 y_2 \geq \frac{x_2}{y_2} + \frac{y_2}{x_2}$$

$$y_1 x_1 + x_1 y_2 - 2 \geq \frac{x_2}{y_2} - \frac{y_2}{x_2} - 2 = \left(\sqrt{\frac{y_2}{x_2}} - \sqrt{\frac{x_2}{y_2}} \right)^2 \geq 0$$

$$(\alpha(1-\alpha) \geq 0) \quad \alpha(1-\alpha)(y_1 x_2 + x_1 y_2) \geq 2\alpha(1-\alpha)$$

$$A \geq \alpha^2 + (1-\alpha)^2 + 2\alpha(1-\alpha) = \alpha^2 + \alpha^2 + 1 - 2\alpha + 2\alpha - 2\alpha^2 = 1$$

~~So~~ $A \geq 1$ we can conclude that $A \in S$. As a consequence S is convex

3) $C = \{x \mid \|x - x_0\|_2 \leq \|x - y\|_2 \text{ for all } y \in S\}$

$$C = \bigcap_{y \in S} \{x \mid \|x - x_0\|_2 \leq \|x - y\|_2\} = \bigcap_{y \in S} C_y$$

Let's say $x \in C_y$ so $\|x - x_0\|_2 \leq \|x - y\|_2 \Leftrightarrow (x - x_0)^T (x - x_0) \leq (x - y)^T (x - y)$

$$\Leftrightarrow x^T x - 2x_0^T x + x_0^T x_0 \leq x^T x - 2y^T x + y^T y$$

$$\Leftrightarrow 2(y - x_0)^T x \leq y^T y - x_0^T x_0$$

So we can write $C_y = \{x \mid \alpha(y-x)^T x \leq y^T y - x_0^T x_0\}$ for $y \in S$

So C_y is a halfspace. Do the intersections of C_y are convex

4) $C = \{x \mid \text{dist}(x, S) \leq \text{dist}(x, T)\}$ where $S, T \subset \mathbb{R}^n$ and $\text{dist}(x, S) = \inf\{\|x - s\|_2 \mid s \in S\}$

C is not convex. because for $n=1$ and $S =]0, 2]$ and $T =]1, 1]$

$$C = \{x \in \mathbb{R} \mid x \in [-\omega, \frac{1}{2}] \cup [\frac{3}{2}, \omega]\} \text{ which is not convex}$$

$$5) S = \{x \mid x + S_2 \subseteq S_1\}$$

$$S = \{x \mid x + y \in S_1, y \in S_2\} = \bigcap_{y \in S_2} \{x \mid x + y \in S_1\} = \bigcap_{y \in S_2} S_1 - y$$

Since $S_1 - y$ is convex for $y \in S_2$, their intersection is also convex

Exercise 2

$$1) f(x_1, x_2) = x_1 x_2 \text{ on } \mathbb{R}_+^2.$$

$$f \text{ is twice differentiable in } \mathbb{R}_+^2 \text{ and } \nabla^2 f(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\det(\nabla^2 f - \lambda \text{Id}) = \det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 - 1 = (\lambda - 1)(\lambda + 1)$$

We can deduce that the eigenvalues of $\nabla^2 f$ are -1 and 1 , so f is not a convex function nor a concave function

Let's $\alpha \in \mathbb{R}$ and $S_\alpha = \{(x_1, x_2) \in \mathbb{R}_+^2 \mid f(x) \geq \alpha\} = \{(x_1, x_2) \in \mathbb{R}_+^2 \mid x_1 x_2 \geq \alpha\}$

Same as in question 2 of exercise n°1 we

have for $\beta \in [0, 1]$; and $(x_1, x_2) \in \mathbb{R}_+^2$ and $(y_1, y_2) \in \mathbb{R}_+^2$

$$(\beta x_1 + (1-\beta)y_1)(\beta x_2 + (1-\beta)y_2) = \beta^2 x_1 x_2 + (1-\beta)^2 y_1 y_2 + \beta(1-\beta)(x_2 y_1 + x_1 y_2)$$

$$\geq \beta^2 \alpha + (1-\beta)^2 \alpha + 2\alpha\beta(1-\beta)$$

$$= \alpha(\beta^2 + (1-\beta)^2 - 2\beta + 1 + 2\beta - 2\beta^2)$$

$$= \alpha$$

So S_α is convex. Conclusion f is quasiconcave

and since $S'_1 = \{(x_1, x_2) \in \mathbb{R}_+^2 \mid x_1 x_2 \leq 1\}$ is not convex so f is not quasiconvex

2) $f(x_1, x_2) = \frac{1}{x_1 x_2}$ on $\mathbb{R}_{++}^2$

f is twice differentiable and $\text{dom}(f) = \mathbb{R}_{++}^2$ convex

and
$$\nabla^2 f = \begin{pmatrix} \frac{2}{x_1^3 x_2} & \frac{1}{x_1^2 x_2^2} \\ \frac{1}{x_1^2 x_2^2} & \frac{2}{x_1 x_2^3} \end{pmatrix} = \frac{1}{x_1^2 x_2^2} \begin{pmatrix} 2 \frac{x_2}{x_1} & 1 \\ 1 & 2 \frac{x_1}{x_2} \end{pmatrix}$$

$$\begin{cases} \det(\nabla^2 f) = \frac{1}{x_1^4 x_2^4} (4 - 1) = \frac{3}{x_1^4 x_2^4} > 0 \Rightarrow \text{the eigenvalues of } \nabla^2 f \text{ have the same sign} \\ \text{tr}(\nabla^2 f) = \frac{2}{x_1^3 x_2} \left(\frac{x_2}{x_1} + \frac{x_1}{x_2} \right) > 0 \Rightarrow \text{the eigenvalues are positive} \end{cases}$$

So the $\nabla^2 f$ is a positive definite matrix conclusion: f is convex on $\mathbb{R}_{++}^2$

→ Let's $\alpha \in \mathbb{R}$ and $S_\alpha = \{x \mid f(x) \leq \alpha\} = \{(x_1, x_2) \in \mathbb{R}_{++}^2 \mid \frac{1}{x_1 x_2} \leq \alpha\}$

for $\alpha \leq 0$ $S_\alpha = \{(x_1, x_2) \in \mathbb{R}_{++}^2 \mid \frac{1}{x_1 x_2} \leq \alpha\} = \emptyset$ which is convex

for $\alpha > 0$ $S_\alpha = \{(x_1, x_2) \in \mathbb{R}_{++}^2 \mid x_1 x_2 \geq \frac{1}{\alpha}\}$ which is a convex set

So S_α is convex set for all α

Conclusion: f is quasiconvex function

3) $f(x_1, x_2) = \frac{x_1}{x_2}$ on $\mathbb{R}_{++}^2$.

here $\text{dom} f = \mathbb{R}_{++}^2$ which is convex and f is twice differentiable on $\mathbb{R}_{++}^2$ and

$$\nabla^2 f = \begin{pmatrix} 0 & -\frac{1}{x_2^2} \\ -\frac{1}{x_2^2} & \frac{2x_1}{x_2^3} \end{pmatrix}$$

$\det \nabla^2 f = \frac{2x_1}{x_2^4} - \frac{1}{x_2^4} < 0$ hence the eigenvalues of $\nabla^2 f$ have different signs

thus $\nabla^2 f \not\geq 0$ and $\nabla^2 f \not\leq 0$ so f is not a convex function nor a concave one

→ Let's $\alpha > 0$ and $S_\alpha = \{x \in \mathbb{R}_{++}^2 \mid f(x) \leq \alpha\} = \{(x_1, x_2) \in \mathbb{R}_{++}^2 \mid x_1 \leq \alpha x_2\}$

Let's $\beta \in [0, 1]$ and $x = (x_1, x_2) \in S_\alpha$ and $y = (y_1, y_2) \in S_\alpha$

$\beta x + (1-\beta)y \in S_\alpha$ because $x_1 \leq \alpha x_2$ and $y_1 \leq \alpha y_2$

so $\beta x_1 + (1-\beta)y_1 \leq \alpha(\beta x_2 + (1-\beta)y_2)$

So S_α is convex conclusion f is quasiconvex

$$S'_\alpha = \{x \in \mathbb{R}_{++}^2 \mid f(x) \geq \alpha\} \\ = \{(x_1, x_2) \in \mathbb{R}_{++}^2 \mid x_1 \geq \alpha x_2\}$$

Let's $\beta \in [0, 1]$: $x = (x_1, x_2) \in S'_\alpha$ and $y = (y_1, y_2) \in S'_\alpha$

So $x_1 \geq \alpha x_2$ and $y_1 \geq \alpha y_2$

$$\beta x_1 + (1-\beta)y_1 \geq \alpha(\beta x_2 + (1-\beta)y_2)$$

So S'_α is convex $\Rightarrow f$ is quasiconvex

Conclusion : f is quasi linear function (quasiconvex and quasiconcave) on $\mathbb{R}_{++}^2$

4) $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ where $0 \leq \alpha \leq 1$ on $\mathbb{R}_{++}^2$

$\{ \text{dom } f = \mathbb{R}_{++}^2 \text{ which is convex} \}$
 $\{ f \text{ is a twice differentiable function} \}$

$$\text{So } \nabla^2 f(x_1, x_2) = \begin{pmatrix} \alpha(\alpha-1)x_1^{\alpha-2}x_2^{1-\alpha} & \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} \\ \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} & \alpha(\alpha-1)x_1^\alpha x_2^{-\alpha-1} \end{pmatrix}$$

$$\det \nabla^2 f(x_1, x_2) = \alpha^2(\alpha-1)^2 x_1^{2\alpha-2} x_2^{-2\alpha} - \alpha^2(1-\alpha)^2 x_1^{2\alpha-2} x_2^{-2\alpha} = 0$$

$$\text{Tr } (\nabla^2 f) = \underbrace{\alpha(\alpha-1)}_{\leq 0} \left(\underbrace{x_1^{\alpha-2} x_2^{1-\alpha}}_{\geq 0} + \underbrace{x_1^\alpha x_2^{-\alpha-1}}_{\geq 0} \right) \leq 0$$

So $\nabla^2 f$ have two eigenvalues one is nul and one is negative so f is concave function

Since f is concave then f is quasiconcave

Exercise 3

1) $f(x) = \text{Tr}(x^{-1})$ on $\text{dom } f = S_{++}^n$

$\text{dom } f = S_{++}^n$ is a convex set.

with $x \in S_{++}^n$ and $v \in S^n$

Let's $g: \mathbb{R} \rightarrow \mathbb{R}$

$t \rightarrow g(t) = f(x + tv)$

if we show that g is convex then f is convex.

Let $t \in \mathbb{R}$ $g(t) = f(x + tv) = \text{Tr}((x + tv)^{-1}) = \text{Tr}(x^{-1}(I + x^{-1/2}vx^{-1/2})^{-1})$

now, the matrix $x^{-1/2}vx^{-1/2}$ is a real symmetric $n \times n$ matrix.

We compute its spectral decomposition: $x^{-1/2}vx^{-1/2} = Q\Lambda Q^T$

where $Q \in \mathbb{R}^{n \times n}$ is orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, with $\lambda_i \in \mathbb{R}$ the eigenvalues of

$$\begin{aligned} x^{-1/2}vx^{-1/2} \quad g(t) &= \text{Tr}(x^{-1}(I + tx^{-1/2}vx^{-1/2})^{-1}) \\ &= \text{Tr}(x^{-1}Q(I + t\Lambda)^{-1}Q^T) \\ &= \text{Tr}(Q^T x^{-1}Q(I + t\Lambda)^{-1}) \\ &= \sum_{i=1}^n (Q^T x^{-1}Q)_{ii} (1 + t\lambda_i)^{-1} \\ &= \sum_{i=1}^n \frac{c_{ii}}{1 + t\lambda_i} \end{aligned}$$

We have $h_i: t \rightarrow \frac{c_{ii}}{1 + t\lambda_i}$ for $i \in \{1, \dots, n\}$ are convex

$$\frac{\partial h_i}{\partial t} = -c_{ii} (1 + t\lambda_i)^{-2} \Rightarrow \frac{\partial h_i}{\partial t} = -2c_{ii} (1 + t\lambda_i)^{-3} \geq 0$$

g is a sum of convex function so it's convex

2) $f(x, y) = y^T x^{-1} y$ on $\text{dom } f = S_{++}^n \times \mathbb{R}^n$

We see that $\frac{1}{2} y^T Q^{-1} y = \sup_x (y^T x - \frac{1}{2} x^T Q x)$

$$\Rightarrow y^T Q^{-1} y = \sup_x (2y^T x - x^T Q x)$$

$$f(x) = \sup_y (g(x, y))$$

g is convex function in (x, y) because it's linear function for x and y

so f which is sup of convex function is convex

3) $f(x) = \sum_{i=1}^n \sigma_i(x)$ and $\sigma_i(x) = \sigma_i(V^T x)$ (since $\sigma_i(V^T x) = \sigma_i(x)$)
 Let's $x \in S^n$ we have the SVD decomposition is $x = U^T D U$ with $U^T U = I$
 where $D = \text{Diag}(\lambda_1, \dots, \lambda_n)$, $S = \text{Diag}(\text{sign}(\lambda_1), \dots, \text{sign}(\lambda_n))$

$$f(x) = \sum_{i=1}^n \sigma_i(x) = \sum_{i=1}^n |\lambda_i| = \sum_{i=1}^n |\text{sign}(\lambda_i) \lambda_i|$$

$$= \text{Tr}(U^T S U D) = \text{Tr}(U^T S U x)$$

U is orthogonal and also D since $S S = \text{Diag}(\text{sign}(\lambda_1)^2, \dots, \text{sign}(\lambda_n)^2) = I = S S$

$Q \rightarrow U^T S Q$ is a permutation of orthogonal matrix
 $\sup_{Q \text{ ortho}} \text{Tr}(Q x) = \sup_{Q \text{ ortho}} \text{Tr}(U^T S Q U x)$

$$= \sup_{Q \text{ ortho}} \text{Tr}(Q U x U^T S) = \sup_{Q \text{ ortho}} (Q |D|)$$

$$\text{Tr}(Q |D|) = \sum_{i=1}^n q_{ii} |\lambda_i| \quad q_{ii} \leq \sqrt{q_{ii}^2} \leq 1$$

orthogonal since Q is orthogonal and $q_{ii} \leq 1$

for $\text{Tr}(Q |D|) \leq f(x)$ so $\sup_{Q \text{ ortho}} \text{Tr}(Q x) \leq f(x)$
 $x \in S^n$ the $f(x) = \sum_{i=1}^n \sigma_i(x) = \sup_{Q \text{ ortho}} \text{Tr}(Q x)$

f is convex since it's the sup of affine function on orthogonal set

Exercise 4: monotone non negative cones K_{m+} $\{x \in \mathbb{R}^n \mid x_1 \geq x_2 \geq \dots \geq x_n \geq 0\}$

1- show K_{m+} is a proper cone:

$\rightarrow K_{m+}$ is closed:

$$K_{m+} = \left(\bigcap_{i=1}^{n-1} \{x_i \geq x_{i+1}\} \right) \cap \{x_n \geq 0\}$$

since K_{m+} is intersection of closed sets then K_{m+} is closed

$\rightarrow K_{m+}$ is solid interior:

Let $x = (x_1, x_2, \dots, x_n) \in K_{m+}$ so $K_{m+} \neq \emptyset \Rightarrow K_{m+}$ has a solid interior

$\rightarrow K$ is pointed (contains no line)

suppose that $x = (x_1, \dots, x_n) \in K_{m+}$

then $x_1, x_2, \dots, x_n = 0 \Rightarrow x_i \geq 0$ for $i \in \{1, n\}$

if K_{m+} contains a line then for $x \in K_{m+}$ we have $-x \in K_{m+}$

But for $x \in K_{m+}$ we have $x_i \geq 0$ so it's impossible to have $-x_i \leq 0$ then $-x \notin K_{m+}$

2. Find the dual cone K_m^*

Dual cone of K_m :

$$K_m^* = \{y \mid y^T x \geq 0 \forall x \in K_m\} = \{y \mid \sum_{i=1}^n y_i x_i \geq 0 \forall x \in K_m\}$$

Let $y \in K_m^*$ $\Rightarrow$ for a given $z_k = \begin{pmatrix} 1 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ k element

it's clear that $z_k \in K_m$; we get

$$y^T z_k = \sum_{i=1}^k y_i \geq 0 \Rightarrow \sum_{i=1}^k y_i \geq 0 \text{ for all } k$$

Reciprocally we have $\sum_{i=1}^k y_i \geq 0 \forall k, 1, \dots, n$

Let's show that $y^T x \geq 0 \forall x \in K_m$: $\sum_{i=1}^n x_i y_i = \sum_{k=1}^{n-1} (x_k - x_{k+1}) \sum_{i=1}^k y_i + x_n \sum_{i=1}^n y_i$ (1)

Let's show the following equality $\sum_{i=1}^n x_i y_i = \sum_{k=1}^{n-1} (x_k - x_{k+1}) \sum_{i=1}^k y_i + x_n \sum_{i=1}^n y_i$

for $n=1$ $\sum_{i=1}^1 (x_i - x_{i+1}) \sum_{i=1}^i y_i + x_{n+1} y_{n+1} = x_1 y_1$

resp. (1) for n and let's show it for $n+1$

$$\begin{aligned} \sum_{i=1}^{n+1} x_i y_i &= \sum_{i=1}^n x_i y_i + x_{n+1} y_{n+1} \\ &= \sum_{k=1}^{n-1} (x_k - x_{k+1}) \sum_{i=1}^k y_i + x_n \sum_{i=1}^n y_i + x_{n+1} y_{n+1} \\ &= \sum_{k=1}^{n-1} (x_k - x_{k+1}) \left(\sum_{i=1}^k y_i + x_{n+1} \left(- \sum_{i=1}^n y_i + \sum_{i=1}^n y_i \right) \right) + x_{n+1} y_{n+1} \\ &= \sum_{k=1}^{n-1} (x_k - x_{k+1}) \sum_{i=1}^k y_i + (x_n - x_{n+1}) \sum_{i=1}^n y_i + x_{n+1} \sum_{i=1}^{n+1} y_i \\ &= \sum_{k=1}^n (x_k - x_{k+1}) \sum_{i=1}^k y_i + x_{n+1} \sum_{i=1}^{n+1} y_i \end{aligned}$$

$$\Rightarrow \text{we have now } \sum_{i=1}^n x_i y_i = y^T x = \underbrace{\sum_{k=1}^n (x_k - x_{k+1})}_{\geq 0} \underbrace{\sum_{i=1}^k y_i}_{\geq 0} + \underbrace{x_{n+1}}_{\geq 0} \underbrace{\sum_{i=1}^{n+1} y_i}_{\geq 0}$$

so $y^T x \geq 0 \forall x \in K_m$

So $K_m^* = \{y \in \mathbb{R}^n \mid \sum_{i=1}^n y_i x_i \geq 0 \forall x \in K_m\}$

$$K_m^* = \left\{ \sum_{i=1}^n y_i \geq 0 \text{ for } k=1, \dots, n \right\}$$

Exercises

1) Max function $f(n) = \max_{i=1, \dots, n} x_i$ on $\mathbb{R}^n$

$$f^*(y) = \max_x y^T x - f(n) = \max_x \left[y^T x - \max_{1 \leq i \leq n} x_i \right]$$

Let's take $y \geq 0$

Suppose $y \geq 0$ then $y^T x - \max_{i=1, \dots, n} x_i \leq \max_{i=1, \dots, n} x_i \left(\sum_{i=1}^n y_i - 1 \right)$

• if $\sum_{i=1}^n y_i > 1$, we can take $x = (p, \dots, p)^T$ and doing $p \rightarrow \infty$ we obtain that

$$y^T x - p \rightarrow +\infty$$

② if $\sum_{i=1}^n y_i < 1$ we do the same with $x = (-p, -p, \dots, -p)^T$ then $y^T x + p \rightarrow -\infty$

③ if $\sum_{i=1}^n y_i = 1$, we have $y^T x - \max_{i=1, \dots, n} x_i \leq 0$

Suppose now that $\exists i \mid y_i < 0$, we can choose x such as

$$x = (0, 0, \dots, 0, -p, 0, \dots, 0)$$

$\uparrow$
index i

if $p \rightarrow +\infty$ we have $y^T x - \max_{i=1, \dots, n} x_i = -p y_i \rightarrow +\infty$

So $f^*(y) = \begin{cases} 0 & \text{if } y \geq 0 \text{ and } \sum_{i=1}^n y_i = 1 \\ +\infty & \text{otherwise} \end{cases}$

2) sum of largest elements. $f(n) = \sum_{i=1}^r x_{[i]}$ on $\mathbb{R}^n$

Suppose y has a negative component, $\exists k$ such as $y_k < 0$

• if we choose x such that $x_k = -t$, $x_i = 0$ for $i \neq k$

so $y \notin \text{dom } f^*$
 $x^T y - f(n) = -t y_k \rightarrow +\infty$

• if y has a component greater than 1, $\exists k$ such as $y_k > 1$. if we choose $x_k = t$, $x_i = 0$ for $i \neq k$

$$x^T y - f(n) = t y_k - t \rightarrow +\infty$$

for $y \in \text{dom } f^*$

if $1^T x \neq r$ we choose $x = t1$ $x^T y - f(n) = t1^T y - tr$ is unbounded as $t \rightarrow -\infty$ or $t \rightarrow +\infty$

if y satisfies all the conditions we have $x^T y \leq f(n)$

for all n we get for $x = 0$ that $f^*(y) = 0$

$$f^*(y) = \begin{cases} 0 & 0 \leq y \leq 1 \text{ and } 1^T y = r \\ +\infty & \text{otherwise} \end{cases}$$

3) Piecewise linear function on $\mathbb{R}$ $g(x) = \max_{i=1, \dots, m} (a_i x + b_i)$ on $\mathbb{R}$.
 The two assumptions tell us that g is a piecewise linear function with vertices $p_i = \frac{b_i - b_{i+1}}{a_{i+1} - a_i}$.

$$g^*(y) = \sup_{x \in \mathbb{R}} \{ yx - \max_{i=1, \dots, m} (a_i x + b_i) \}$$

If $y < a_1$ or $y > a_m$, we have that $yx - \max_{i=1, \dots, m} (a_i x + b_i)$ does not have a supremum.

Suppose now that $a_1 \leq y \leq a_m$. Then, as we suppose that a_i are sorted in increasing order, we have that $\exists k$ $a_k \leq y \leq a_{k+1}$. Then $y - a_i \geq 0 \forall i=1, \dots, k$ and $y - a_i \leq 0 \forall i=k+1, \dots, m$. As a consequence, the supremum of $yx - \max_{i=1, \dots, m} (a_i x + b_i)$ is reached at p_k .

$$\begin{aligned} \text{Pic. we obtain } g^*(y) &= g(p_k) = a_k p_k + b_k = y \frac{b_k - b_{k+1}}{a_{k+1} - a_k} - a_k \frac{b_k - b_{k+1}}{a_{k+1} - a_k} + b_k \\ &= (y - a_k) \left(\frac{b_k - b_{k+1}}{a_{k+1} - a_k} \right) + b_k \end{aligned}$$

In light of the above, we conclude that:

$$g^*(y) = \begin{cases} (y - a_k) \frac{b_k - b_{k+1}}{a_{k+1} - a_k} + b_k & \text{if } a_1 \leq y \leq a_m \\ +\infty & \text{otherwise} \end{cases}$$